

Decoupling: Most (but not all) of the details of very small-distance phenomena tend to be largely irrelevant for the description of much larger systems

FFT: Mathematical tools for exploiting decoupling

An illustrative Toy Model:

*To be of interest it must have two types of particles which interact in a simple/nontrivial way.

- One particle (must) be much more massive
- ϕ : Scalar field (Complex)

$$S := - \int d^4x [\partial_\mu \phi^* \partial^\mu \phi + V(\phi, \phi^*)]$$

Self-interaction

$$V(\phi; \phi^*) = \frac{\lambda}{4} (\phi^* \phi - v^2)^2, \quad \lambda, v^2 \in \mathbb{R}^+$$

min: $|\phi| = v$



Basics of Scattering Theory:

- Only need changes in $(E, \vec{p}) \rightarrow \Delta P^m$
- Can approximate the real states using a class of energy eigenstates

$$H_{\text{tot}} = H_0 + V$$

H_0 : describes the evolution of $|i\rangle$ & $|f\rangle$

$$\begin{cases} i\partial_t |\psi\rangle = \hat{H}_0 |\psi\rangle \\ \Rightarrow |f\rangle \approx e^{-iH_0 t} |i\rangle \end{cases}$$

- Energy eigenstates of H : $|\alpha\rangle$

$$H_0|\alpha\rangle = E_\alpha|\alpha\rangle$$

- A wave packet of such states:

$$|\phi_f\rangle := \int d\alpha f(\alpha)|\alpha\rangle$$

$f(\alpha)$: Normalizable packet

α : Contains continuum-normalized momentum-states

Set of
eigenstates
of H_0



Set of
eigenstates
of H

$$H|\alpha\rangle = E_\alpha|\alpha\rangle \quad |\alpha\rangle \equiv \uparrow^e$$

$$\lim_{t \rightarrow \infty} e^{-iHt} |\phi_f\rangle_0 = \lim_{t \rightarrow \infty} e^{-iH_0 t} |\phi_f\rangle$$

$$\lim_{t \rightarrow -\infty} e^{-iHt} |\phi_f\rangle_i = \lim_{t \rightarrow -\infty} e^{-iH_0 t} |\phi_f\rangle$$

$$\lim_{t \rightarrow \infty} e^{-iHt} |\alpha\rangle_0 = \lim_{t \rightarrow \infty} e^{-iH_0 t} |\alpha\rangle \quad \left| \lim_{t \rightarrow -\infty} e^{-iHt} |\alpha\rangle_i = \lim_{t \rightarrow -\infty} e^{-iH_0 t} |\alpha\rangle \right|$$

* Matrix of all possible amplitudes between
scattering energy eigenstates:

$$S_{\beta\alpha} \equiv \langle\beta|\alpha\rangle \quad (S\text{-matrix})$$

Scattering operator: $\hat{S}$

- Defined as that op. whose matrix elements between H_0 eigenstates $|\alpha\rangle$ reproduce the above:

$$\langle\beta|\hat{S}|\alpha\rangle := S_{\beta\alpha}$$

Møller wave operator:

$$\Omega(t) := e^{iHt} e^{-iH_0 t}$$

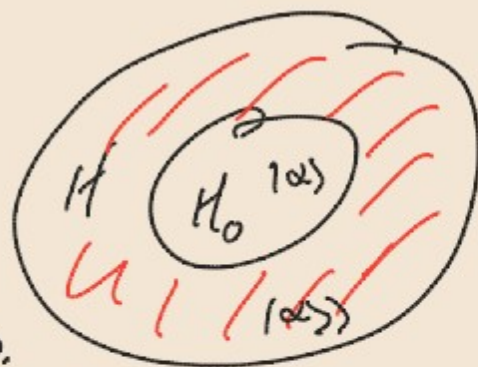
because

$$|\alpha\rangle\rangle_+ = \lim_{t \rightarrow \infty} \Omega(t) |\alpha\rangle \quad \text{and} \quad |\alpha\rangle\rangle_- = \lim_{t \rightarrow -\infty} \Omega(t) |\alpha\rangle$$

$\Omega(t) \sim e^{iVt}$ however not exactly since we must use Baker-Campbell-Hausdorff.

Define: $\Omega^\pm := \lim_{t \rightarrow \pm\infty} \Omega(t)$

- $\Omega^\pm$ are not unitary in the presence of 'Bound states' contained in the spectrum of H but not H_0



The S -matrix^{op.} is then given by
 $\hat{S} = \lim_{t \rightarrow \infty} \lim_{t' \rightarrow -\infty} \Omega^*(t) \Omega(t') = (\Omega^+)^* \Omega^-$

Semiclassical Spectrum: $\lambda \ll 1$
and both v and $|p|$ are $\mathcal{O}(\lambda^{1/2})$

